

Testing higher-order theories of visual awareness using priming

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Two competing classes of theories posit that different neural representations are necessary for “phenomenal awareness,” the subjective, first-person experience we have of seeing something, like a red apple. First-order theories assert that first-order representations in sensory brain regions are sufficient for phenomenal awareness, whereas higher-order theories (HOTs) assert that higher-order representations tracking first-order ones are necessary. Priming occurs when exposure to a stimulus facilitates subsequent processing of similar stimuli. According to HOTs, priming higher-order neural representations will change awareness reports but not performance. Thus, this study aims to determine whether priming of awareness reports can be obtained in the absence of performance changes.

Observers performed a simultaneous orientation detection and discrimination task on a grating target that appeared on half of the trials. The grating target was tilted ± 45 degrees and presented foveally at threshold and near-threshold contrasts. Before the target, observers were exposed to either a “high awareness” or “low awareness” condition, in which a stream (60 s initial exposure with 10 s top-ups) of peripheral gratings was presented at mostly visible or not visible contrasts, respectively.

Data collection is ongoing ($n=8$). Our preliminary results suggest detection criterion is lower following high vs. low awareness exposure while perceptual sensitivity for discrimination and detection remains unchanged. This priming effect is specific to subjective awareness reports. Thus, our preliminary results support HOT predictions that priming higher-order representations could change awareness reports without changing performance.

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